MATH 420-02

Least Squares and Minimizing Distance

Theorem:

Let $y \in V$ and let W be a subspace of V. If $y = w + u$, where $w \in W$ and $u \in W^{\perp}$, then w is the closet point in W to y.

Proof:

Let $x \in W$ and compare $\| y - x \|^2$ to $\| y - w \|^2$.

$\| y - x \|^2 = < y - x, y - x > \; = \; < y, y > - < y, x > - < x, y > + < x, x >$
$= \; < w + u, w + u > - < w + u, x > - < x, w + u > + < x, x >$
$= \; < w, w > + < w, u > + < u, w > + < u, u > - < w, x > - < u, x > - < x, w > - < x, u > + < x, x >$
$= \; < w, w > +0 + 0 + < u, u > - < w, x > - < x, w > + < x, x >$
$= \; < w - x, w - x > + < u, u > \; = \; \| w - x \|^2 + \| u \|^2$

$\| y - w \|^2 = \; \| w + u - w \|^2 = \; \| u \|^2$

Thus $\| y - x \| \; \geq \; \| y - w \|$

Definition:

Let $\| \;\; \|$ be the norm on $\mathbb{R}^n$ defined by $\| x \| = \sqrt{x^T x}$. If $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear operator and b is in $\mathbb{R}^n$ a **least-squares solution** of $T(x) = b$ is an $\hat{x}$ in $\mathbb{R}^n$ such that

$$\left\| b - T(\hat{x}) \right\| \leq \| b - T(x) \|$$

for all x in $\mathbb{R}^n$.

Lemma:

Let $T : V \to V$ be a linear operator. Then Rank (T) = Rank$(T^* T)$ = Rank(T^*) and hence $R(T^*) = R(T^* T)$.

Proof:

Although the result is about the rank, we will approach the proof by working with the nullity. Let $x \in N(T)$. Then $T^* T(x) = T^*(0) = 0$, so $x \in N(T^* T)$. Conversely, let $x \in N(T^* T)$.

$0 = \; < 0, x > \; = \; < T^* T(x), x > \; = \; < T(x), \; T^{**}(x) > \; = \; < T(x), \; T(x) >$, so $T(x) = 0$ so $x \in N(T^* T)$.

Thus $N(T) = N(T^* T)$, so nullity(T) = nullity $(T^* T)$ and hence rank(T) = rank$(T^* T)$. But we know rank(T^*) = rank(T).

Notice that if $y \in R(T^* T)$, there exists an x so that $y = T^* T(x) = T^*(T(x))$ so $y \in R(T^*)$. Thus $R(T^* T) \subseteq R(T^*)$ and since they have the same dimension, they must be equal.

Theorem:

Let $T : V \to V$ be a linear operator and let $b \in V$, where dim $(V) = n$. (We want to solve $T(x) = b$, but there might not be a solution.) Then there exists $u \in V$ such that $T^* T\,(u) = T^* b$ and $\| T(u) - b \| \; \leq \; \| T(x) - b \|$ for all $x \in V$. Moreover, if Rank $(T) = n$, then $u = (T^* T)^{-1} T^*(\, b)$.

Proof:

Since $R(T^* T) = R(T^*)$ and $T^*(b) \in R(T^*)$, there must be an element $u \in V$ such that $T^* T(u) = T^*(b)$.

We will establish that $T(u)$ is the closest point to b in $R(T)$ by observing that

$$b = T(u) + (b - T(u))$$

Where $T(u) \in R(T)$ and $b - T(u)$ is in $R(T)^{\perp}$.

Let $y \in R(T)$. Write $y = T(x)$ for some $x \in V$. Then

$< b - T(u), y > = \; < b - T(u), \; T(x) >$
$= \; < T^*(b - T(u)), x > = \; < T^*(b) - T^*(T(u)), x > = \; < 0, x > = 0$

Thus our decomposition above of b into an element in $R(T) + R(T)^{\perp}$ is correct. Thus $T(u)$ is a closest point to b in V.

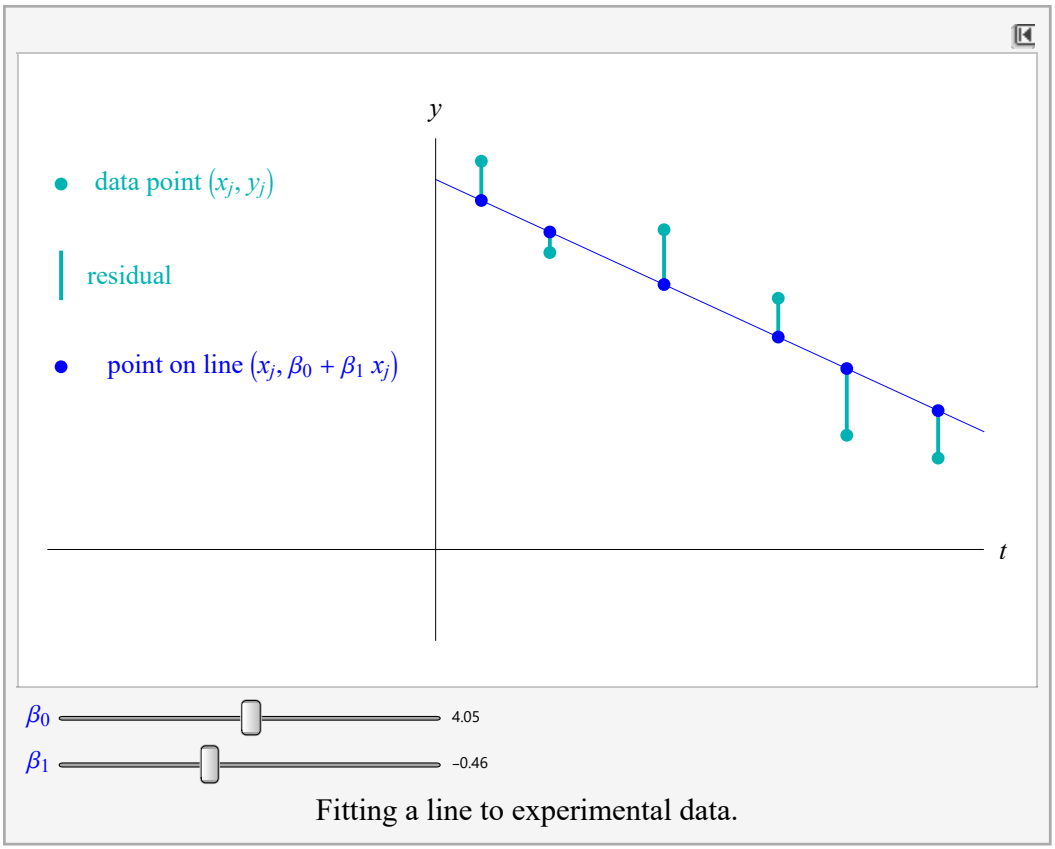

FIGURE 1 ◆

Definition:

The **least-squares line** is the line $y = \beta_0 + \beta_1 x$ that minimizes the sum of the squares of the residuals. This line is also called a **line of regression of y on x**, because any errors in the data are assumed to be only in the y-coordinates. The coefficients β_0, β_1 of the line are called (linear) **regression coefficients**.

Predicted y−value		Observed y−value
$\beta_0 + \beta_1 x_1$	=	y_1
$\beta_0 + \beta_1 x_2$	=	y_2
$\vdots$		$\vdots$
$\beta_0 + \beta_1 x_n$	=	y_n

TABLE A

$$X\boldsymbol{\beta} = \mathbf{y}, \text{ where } X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \ \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \ \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \tag{1}$$

Remark:

Data points rarely lie on a line. Computing the least-squares solution of $X\boldsymbol{\beta} = \mathbf{y}$ is equivalent to finding the $\boldsymbol{\beta}$ that determines the least-squares line in Figure 1.

EXAMPLE:

Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the data points (2, 1), (5, 2), (7, 3), and (8, 3).

SOLUTION:

Set up the matrix equation $X \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = y$ where:

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ 1 & x_4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 3 \end{pmatrix}$$

Solve $X^* X \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = X^* y$

In[3]:= $\text{Transpose}\left[\begin{pmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{pmatrix}\right]$

Out[3]= $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{pmatrix}$

In[2]:= $\$\text{PrePrint} = \text{TraditionalForm}; \text{Transpose}\left[\begin{pmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{pmatrix}\right] . \begin{pmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{pmatrix}$

Out[2]= $\begin{pmatrix} 4 & 22 \\ 22 & 142 \end{pmatrix}$

In[ ]:= $\text{Transpose}\left[\begin{pmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{pmatrix}\right] . \begin{pmatrix} 1 \\ 2 \\ 3 \\ 3 \end{pmatrix}$

Out[ ]= $\begin{pmatrix} 9 \\ 57 \end{pmatrix}$

In[ ]:= $\text{RowReduce}\left[\begin{pmatrix} 4 & 22 & 9 \\ 22 & 142 & 57 \end{pmatrix}\right]$

Out[ ]= $\begin{pmatrix} 1 & 0 & \frac{2}{7} \\ 0 & 1 & \frac{5}{14} \end{pmatrix}$

The least squares best fit line is $y = \frac{2}{7} + \frac{5}{14} x$

SUMMARY:

Solve $X^* X \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = X^* y$ by solving $\begin{pmatrix} 4 & 22 \\ 22 & 142 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 9 \\ 57 \end{pmatrix}$

$\text{RowReduce}\left[\begin{pmatrix} 4 & 22 & 9 \\ 22 & 142 & 57 \end{pmatrix}\right]$ to get $\begin{pmatrix} 1 & 0 & \frac{2}{7} \\ 0 & 1 & \frac{5}{14} \end{pmatrix}$.

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EXAMPLE:

If a machine "learns" from the points (2, 1), (5, 2), (7, 3), and (8, 3) by creating a least squares line, what outcome will it predict for the inputs 4 and 6?

In[ ]:= $y = \frac{2}{7} + \frac{5}{14} x;\ x = 4;\ y$

Out[ ]= $\frac{12}{7}$

In[ ]:= `Clear[x]; y = 2/7 + 5/14 x; x = 6; y`

Out[ ]= $\frac{17}{7}$

Least-Squares Fitting of Other Curves:

When data points $(x_1, y_1), \ldots, (x_n, y_n)$ on a scatter plot do not lie close to any line, it may be appropriate to postulate some other functional relationship between x and y.

$$y = \beta_0 f_0(x) + \beta_1 f_1(x) + \cdots + \beta_k f_k(x) \tag{2}$$

where $f_0, \ldots, f_k$ are known functions and $\beta_0, \ldots, \beta_k$ are parameters that must be determined.

EXAMPLE:

Suppose data points $(x_1, y_1), \ldots, (x_n, y_n)$ appear to lie along some sort of parabola instead of a straight line.

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 \tag{3}$$

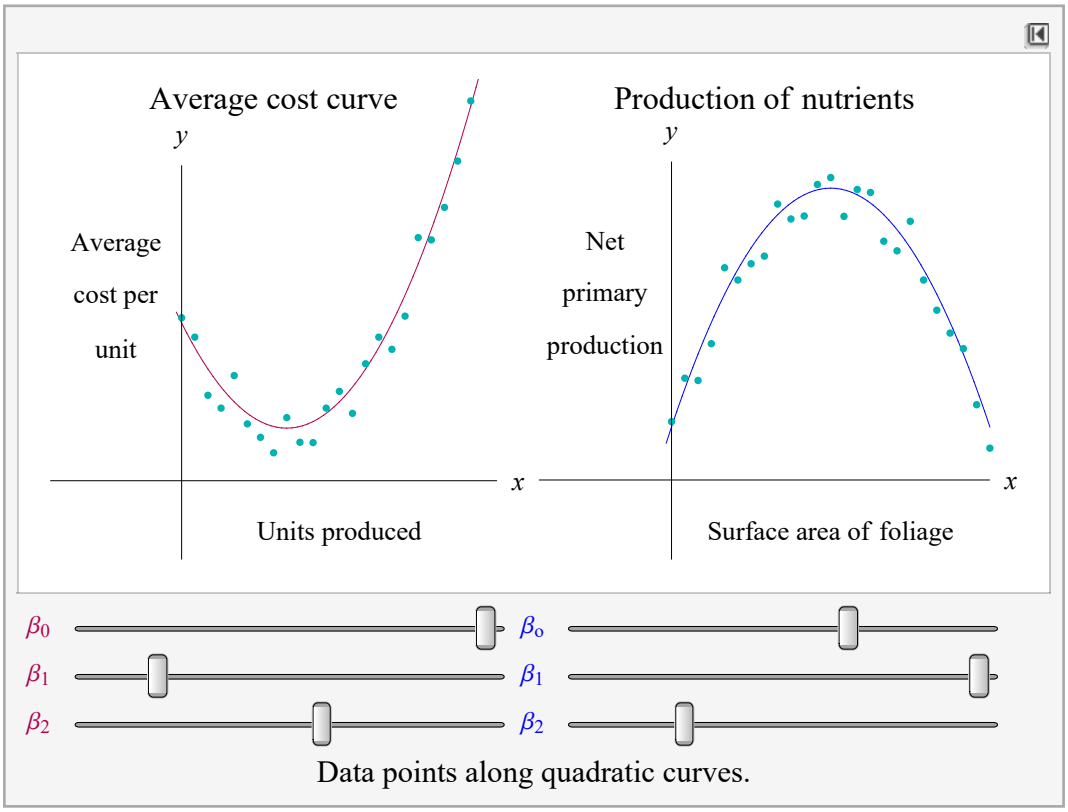

FIGURE 4-5 ◆

SOLUTION:

Suppose the actual values of the parameters are $\beta_0, \beta_1, \beta_2$. Then the coordinates of the first data point (x_1, y_1) satisfy an equation of the form

$$y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon_1$$

where ϵ_1 is the residual error between the observed value y_1 and the predicted y-value $\beta_0 + \beta_1 x_1 + \beta_2 x_1^2$. Each data point determines a similar equation:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \epsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + \epsilon_n \end{aligned}$$

It is a simple matter to write this system of equations in the form $\mathbf{y} = X\boldsymbol{\beta} + \epsilon$. To find X, inspect the first few rows of the system and look for the pattern.

$$\begin{array}{ccccccc} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} & = & \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix} & \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} & + & \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \\ \mathbf{y} & = & X & \boldsymbol{\beta} & + & \boldsymbol{\epsilon} \end{array}$$

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EXAMPLE:

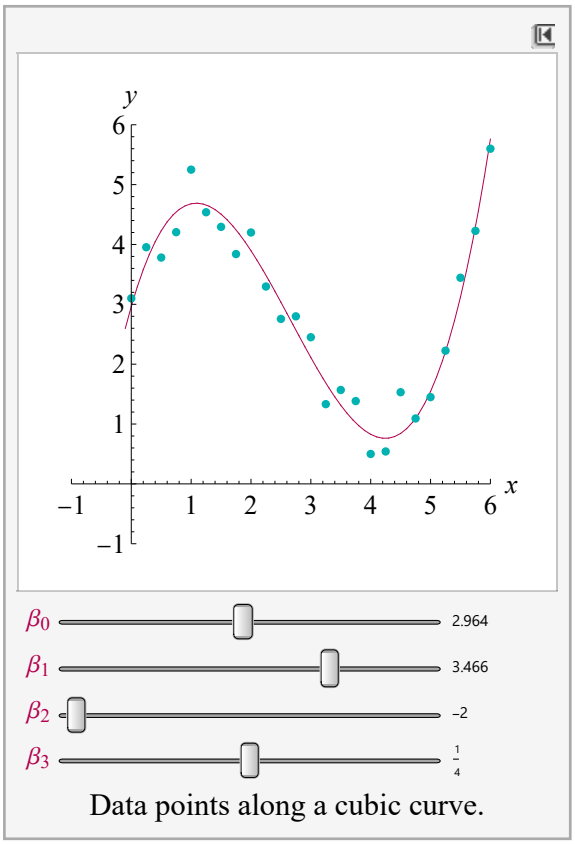

FIGURE 6 ◆

If data points tend to follow a pattern such as in the figure above, then an appropriate model might be an equation of the form

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

SOLUTION:

By an analysis similar to that in Example 2, we obtain

$$\begin{array}{cccc} \text{Observation vector} & \text{Design matrix} & \text{Parametric vector} & \text{Residual vector} \\ \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, & X = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 \end{bmatrix}, & \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}, & \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \end{array}$$

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Multiple Regression

Suppose an experiment involves two independent variables—say, u and v—and one dependent variable, y. A simple equation for predicting y from u and v has the form

$$y = \beta_0 + \beta_1 u + \beta_2 v \tag{4}$$

A more general prediction equation might have the form

$$y = \beta_0 + \beta_1 u + \beta_2 v + \beta_3 u^2 + \beta_4 uv + \beta_5 v^2 \tag{5}$$

or in general

$$y = \beta_0 f_0(u, v) + \beta_1 f_1(u, v) + \cdots + \beta_k f_k(u, v)$$

with $f_0, \ldots, f_k$ any sort of known functions and $\beta_0, \beta_1, \ldots, \beta_k$ unknown weights.

EXAMPLE:

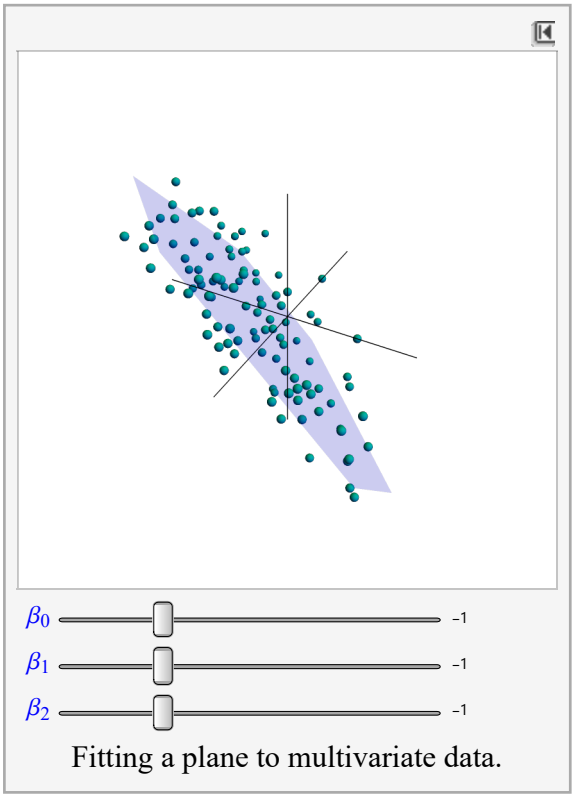

FIGURE 7 ◆

SOLUTION:

We expect the data to satisfy the following equations:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 u_1 + \beta_2 v_1 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 u_2 + \beta_2 v_2 + \epsilon_2 \\ \vdots & \qquad\qquad \vdots \\ y_n &= \beta_0 + \beta_1 u_n + \beta_2 v_n + \epsilon_n \end{aligned}$$

This system has the matrix form $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where

$$\begin{array}{cccc} \text{Observation vector} & \text{Design matrix} & \text{Parametric vector} & \text{Residual vector} \\ \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, & X = \begin{bmatrix} 1 & u_1 & v_1 \\ 1 & u_2 & v_2 \\ \vdots & \vdots & \vdots \\ 1 & u_n & v_n \end{bmatrix}, & \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}, & \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \end{array}$$

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In each of these scenarios, we solve $X^* X\beta = X^* y$ to find the "best" choices for β.